

Data Visualization

1-Introduction

- Used 4 datasets: Iris, Tips, Flights, Custom Monthly Sales.
- Goal: visually explore the data and extract meaningful insights.

2-Findings

Custom Monthly Sales Dataset:

- Sales gradually increase from January to mid-year.
- Peak sales occur in July and August (summer months).
- Sales decrease slightly towards the end of the year.

Tips Dataset:

- Total bills are higher on certain days (e.g., Thursday & Saturday) → weekend effect.
- Differences in tips between male and female customers are minor, but some outliers exist.
- Customers tend to tip more when the total bill is higher.

Flights Dataset:

- The number of passengers steadily increases from 1949 to 1960 → clear upward trend.
- Summer months (June–August) consistently have the highest passenger counts.
- Winter months (Jan–Feb, Nov–Dec) are the lowest in passenger traffic.

Iris Dataset:

- Petal length & width clearly separate setosa from other species.
- Versicolor and Virginica overlap slightly but can still be distinguished using petal dimensions.
- Sepal length & width are less distinct but still show patterns between species.

3-Methodology

- Used **Seaborn + Matplotlib** for plotting.
- Applied style 'darkgrid' and palette 'pastel' for clarity.
- Used lineplot, kdeplot, boxplot, heatmap, and Subplots for organized visualization.

4-Plotly

- Plotly is a library for **interactive data visualization**.
- Allows zooming, hovering, and exporting charts.
- Useful for dashboards and interactive reports.

5-Matplotlib vs Seaborn

Aspect	Matplotlib	Seaborn
Ease of use	Flexible, verbose	High-level, simpler
Styling	Needs manual work	Beautiful default styles
Statistical plots	Limited	Built-in support (box, violin, kde)
When to use	Custom, precise plots	Quick, nice-looking charts